

**User**

"Locate the handle of the mug."

phrases $\phi(x)$:
"the handle of the mug"
"the mug"

**Output**



mask $m_{v^\star}$ ("handle")

Qwen3-VL Vision Encoder

**SAM 3**
32×32 point grid (query-free)

**SAM 3 text**
top-3 instances per phrase

**Query-aware candidate mask pool**



grid nodes (dedup 0.9, ≤256)




text instances (dedup 0.95)

**Containment tree**

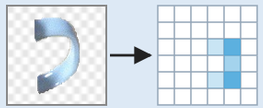
$\text{parent}(v) = \arg\min_u |m_u|$
s.t. $|m_u \cap m_v| \geq 0.8\,|m_v|$



Mask Tokenizer

Text Tokenizer

## Mask Tokenization




appearance
$p_v = \sum_j \tilde{c}_{v,j}\, e_j$

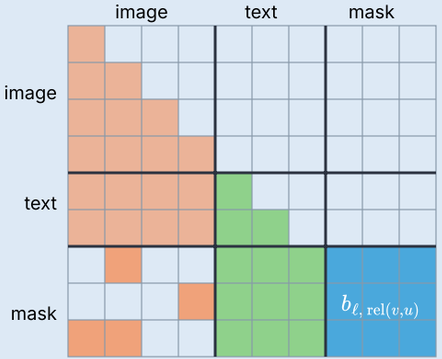
geometry
$g_v \in \mathbb{R}^9$

MLP + RMS norm

$\hat{x}_v$

mRoPE position $(\bar{r}_v, \bar{c}_v)$ = centroid of $m_v$

## Region-restricted attention




- mask token: query text + its own region (mask dilated by 1 cell) + other masks
- mask–mask bias per layer $\ell$ and tree relation (parent, child, sibling, ...)
- text and image tokens never see mask tokens

**Selection head** $s_v = \lambda \cos\left(\psi_q(h_{\text{SEL}}),\, \psi_k(z_v)\right) + \beta, \qquad v^\star = \arg\max_v\, s_v$

## Large Language Model

Qwen3-VL-4B + LoRA (r = 16)



Visual tokens $e_j$



Candidate mask tokens $\hat{x}_v$ (one per mask)




Query + answer "... <SEL>"




Image + query (causal)



Mask tokens (appended, one pass)